



21BTCS503 –Introduction to Data Science

Class - T.Y. (SEM-V), Core

Unit - II Data Science and Big data

AY 2024-2025



Unit II – Data Science and Big data

09 hours

- Defining data science and big data, Recognizing the different types of data, Gaining insight into the data science process, Data Science Process: Overview, Different steps, Machine Learning Definition and Relation with Data Science





Data Science and Big Data

Data Science and Big Data are interconnected fields that deal with the collection, analysis, and interpretation of large volumes of data. Here are some key points and concepts:

- **Data Science**
- **Definition:** Data Science is a multidisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data. It encompasses a variety of techniques from statistics, machine learning, data mining, and big data technologies.
- **Key Components:**
- **Data Collection:** Gathering data from various sources.
- **Data Preparation:** Cleaning and organizing data for analysis.
- **Data Exploration:** Analyzing data to discover patterns and insights.
- **Model Building:** Creating algorithms and models to make predictions or classifications.
- **Model Evaluation:** Assessing the accuracy and performance of models.
- **Presentation:** Communicating findings through visualizations and reports.
- **Automation:** Implementing automated systems for continuous data analysis.



- **Applications:**
 - Predictive analytics
 - Recommender systems
 - Fraud detection
 - Customer segmentation
 - Sentiment analysis

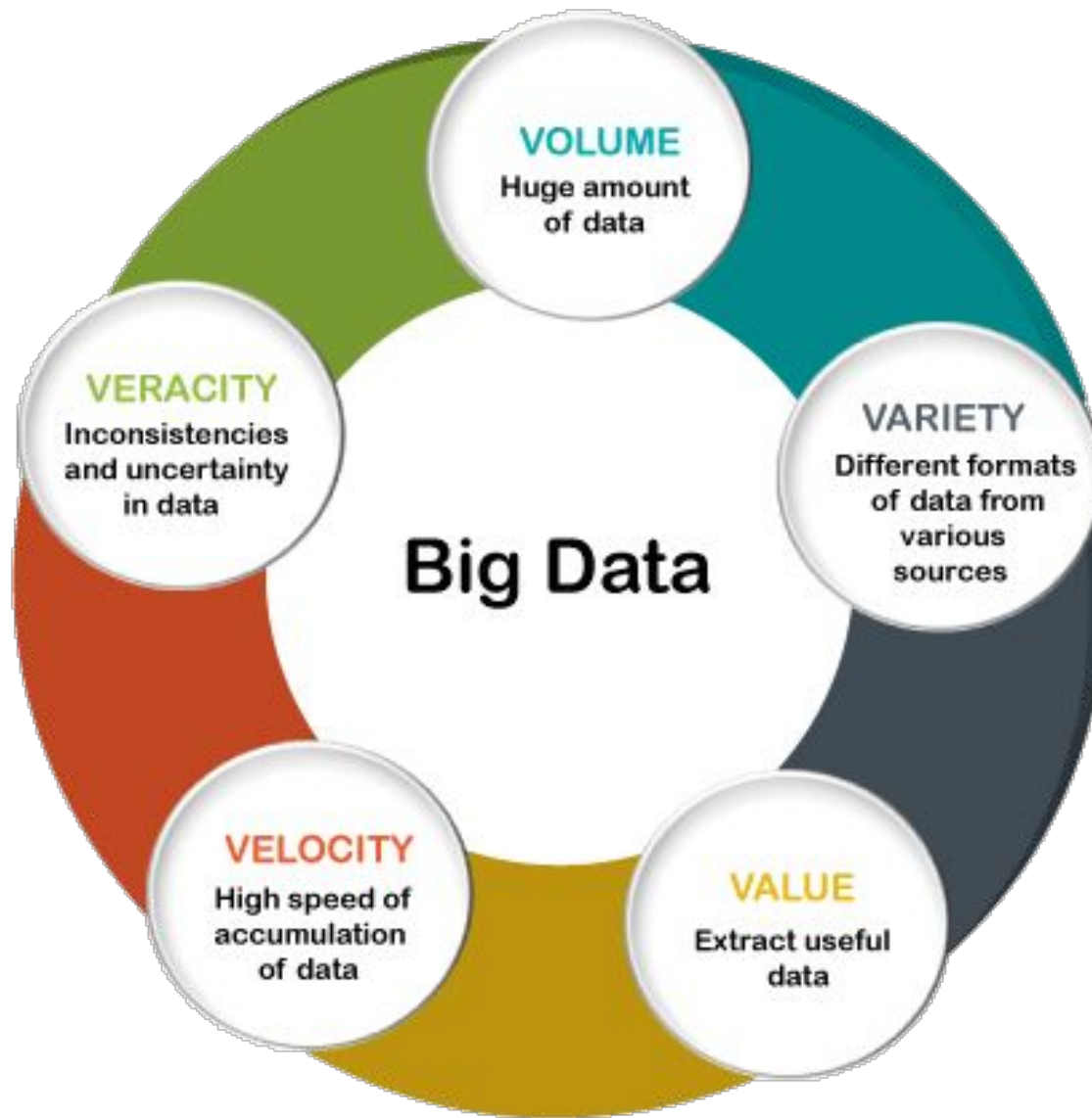




Big Data

- **Definition:** Big Data refers to extremely large data sets that may be analyzed computationally to reveal patterns, trends, and associations, especially relating to human behavior and interactions.
- **Characteristics (The 6 Vs):**
 1. **Volume:** The quantity of data generated and stored.
 2. **Velocity:** The speed at which data is generated and processed.
 3. **Variety:** The different types of data (structured, unstructured, semi-structured).
 4. **Veracity:** The uncertainty or trustworthiness of data.
 5. **Value:** The potential insights and benefits derived from analyzing big data.
 6. **Variability:** How often does the meaning or shape of your data change?
- **Technologies and Tools:**
 1. Hadoop
 2. Spark
 3. NoSQL databases (e.g., MongoDB, Cassandra)
 4. Distributed storage systems (e.g., HDFS)
 5. Data lakes







- **Applications:**

1. Real-time analytics
2. Internet of Things (IoT)
3. Social media analysis
4. Healthcare data analysis
5. Supply chain management

- **Relation between Data Science and Big Data**

Data Science leverages Big Data technologies to handle and analyze vast amounts of data that traditional data processing tools cannot manage efficiently. Big Data provides the raw material, while Data Science provides the methods and tools to turn this raw material into actionable insights.

- **Gaining Insight into the Data Science Process**

Understanding the data science process involves recognizing the different steps involved from data collection to final presentation. Each step requires specific skills and tools, and mastering these can lead to successful data-driven decision-making.



Steps in the Data Science Process

Defining the Research Goal: Understanding the problem to be solved.

Retrieving Data: Collecting data from various sources.

Data Preparation: Cleaning and transforming data.

Data Exploration: Performing initial investigations on data.

Model Building: Developing algorithms to solve the problem.

Evaluation: Testing and validating the models.

Presentation: Sharing insights with stakeholders.

Automation: Creating automated systems for continuous analysis.





Machine Learning and Data Science

Machine learning is a subset of data science that focuses on building systems that can learn from data. It involves training algorithms on data so that they can make predictions or classifications without being explicitly programmed to perform the task.

Types of Machine Learning:

Supervised Learning: Learning from labeled data (e.g., classification, regression).

Unsupervised Learning: Learning from unlabeled data (e.g., clustering, association).

Reinforcement Learning: Learning by interacting with the environment to maximize some notion of cumulative reward.

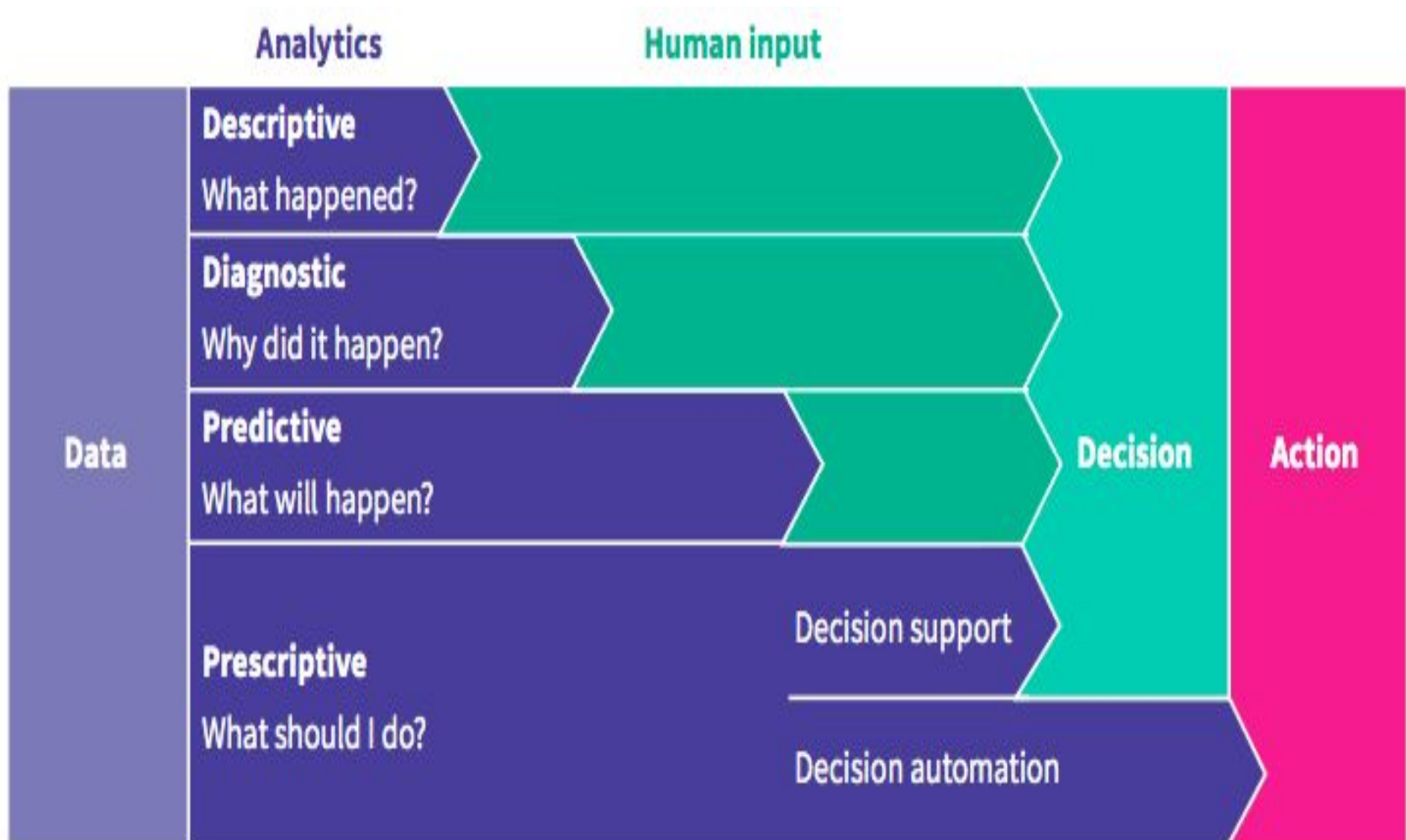
Key Takeaways

Data Science and Big Data are closely related fields essential for modern data analysis.

The data science process involves multiple steps, each crucial for deriving meaningful insights.

Big Data technologies enable the handling of vast amounts of data, which data science techniques can analyze.

Machine learning is a critical component of data science, allowing for predictive and prescriptive analytics.





Recognizing the different types of data :

Recognizing different types of data involves understanding and distinguishing between various forms and sources of data. Here's a detailed breakdown:

Types of Data

1. Structured Data

- **Definition:** Data that is highly organized and easily searchable. It is often stored in relational databases and spreadsheets.
- **Examples:** Tables in a database, Excel spreadsheets, CSV files.
- **Characteristics:** Defined data types, clear relationships, and fixed schema.

2. Unstructured Data

- **Definition:** Data that does not have a predefined data model or is not organized in a predefined manner.
- **Examples:** Text documents, emails, social media posts, videos, images.
- **Characteristics:** Lacks a specific format or structure, more complex to analyze and manage.

3. Semi-Structured Data

- **Definition:** Data that does not conform to a rigid structure but contains tags or markers to separate data elements.
- **Examples:** JSON, XML, NoSQL databases.
- **Characteristics:** Flexible schema, self-describing, often used in web data exchanges.



Sources and Formats of Data

1. Natural Language Data

Definition: Data in the form of human language, typically text or speech.

Examples: Books, articles, conversations, tweets.

Characteristics: Requires natural language processing (NLP) for analysis.

2. Machine-Generated Data

Definition: Data created by machines without human intervention.

Examples: Sensor data, logs from servers, data from IoT devices.

Characteristics: Often high-volume, continuous, and precise.

3. Graph-Based or Network Data

Definition: Data that represents relationships and connections between entities.

Examples: Social networks, knowledge graphs, citation networks.

Characteristics: Nodes and edges, used in network analysis.



4. Audio Data

Definition: Data in the form of sound.

Examples: Music files, recordings, voice commands.

Characteristics: Analyzed using audio signal processing.

5. Image and Video Data

Definition: Visual data in the form of still images or moving pictures.

Characteristics: Requires image processing and computer vision techniques.

6. Streaming Data

Definition: Data that is continuously generated and delivered in real-time.

Examples: Live feeds, real-time analytics, stock market data.

Characteristics: High velocity, often needs real-time processing and analytics

Examples: Photographs, video clips, CCTV footage.





Gaining insight into the data science process :

Gaining insight into the data science process involves understanding the key steps and methodologies used to extract meaningful information and knowledge from data. Here's a comprehensive overview of the data science process:

1. Defining the Problem and Setting the Research Goal

Objective: Clearly define the problem you aim to solve or the question you want to answer.

Key Activities:

Understand the business context and objectives.

Identify the specific goals of the data science project.

Formulate research questions or hypotheses.

2. Retrieving Data

Objective: Collect relevant data from various sources.

Key Activities:

Identify data sources (databases, APIs, web scraping, surveys, etc.).

Extract data using appropriate methods and tools.

Ensure data is collected in a format suitable for analysis.





3. Data Preparation

Objective: Clean and preprocess the data to make it suitable for analysis.

Key Activities:

Handle missing values, outliers, and inconsistencies.

Convert data types and format data as needed.

Normalize or standardize data.

Feature engineering: create new features or variables that may be useful for analysis.

4. Data Exploration

Objective: Perform initial analysis to understand the data and identify patterns.

Key Activities:

Use descriptive statistics to summarize the data.

Visualize data using charts, graphs, and plots.

Identify correlations and relationships between variables.

Perform exploratory data analysis (EDA) to generate insights and hypotheses.





5. Model Building

Objective: Develop predictive or descriptive models using statistical or machine learning techniques.

Key Activities:

Choose appropriate modeling techniques (e.g., regression, classification, clustering).

Split data into training and testing sets.

Train models on the training data.

Evaluate model performance using appropriate metrics (e.g., accuracy, precision, recall).

Tune hyperparameters to optimize model performance.

6. Model Evaluation and Validation

Objective: Assess the model's performance and validate its accuracy.

Key Activities:

Use cross-validation techniques to ensure model robustness.

Compare model performance with baseline models.

Validate the model using unseen test data.

Ensure the model generalizes well to new data.

7. Presentation and Visualization

Objective: Communicate the findings and insights effectively to stakeholders.

Key Activities:

Create visualizations to highlight key insights and trends.

Use storytelling techniques to explain the results.

Prepare reports and presentations that summarize the analysis and findings.

Provide actionable recommendations based on the insights.





8. Deployment and Automation

Objective: Implement the model in a production environment and automate the process.

Key Activities:

Deploy the model to production systems for real-time or batch processing.

Monitor model performance and update it as necessary.

Automate data collection, preprocessing, and analysis pipelines.

Ensure the system is scalable and can handle large volumes of data.

9. Monitoring and Maintenance

Objective: Continuously monitor and maintain the deployed models.

Key Activities:

Track model performance over time to detect any degradation.

Retrain models with new data to keep them up-to-date.

Implement alert systems to notify of any issues or anomalies.

Regularly review and update data pipelines and workflows.

10. Documentation and Collaboration

Objective: Maintain comprehensive documentation and foster collaboration.

Key Activities:

Document data sources, methodologies, code, and findings.

Use version control systems to manage code and data changes.

Collaborate with team members and stakeholders using collaborative tools.

Share knowledge and insights through workshops, seminars, and publications.





Machine Learning Definition

Machine Learning (ML) is a subset of artificial intelligence (AI) that focuses on developing algorithms and statistical models that enable computers to learn from and make predictions or decisions based on data. Instead of being explicitly programmed to perform a task, machine learning models are trained on data, which allows them to identify patterns and make informed decisions.

Key Characteristics of Machine Learning

- **Data-Driven:** ML relies heavily on data to train models and make predictions.
- **Pattern Recognition:** ML algorithms detect patterns and relationships in data.
- **Adaptability:** Models can improve over time as they are exposed to more data.
- **Automation:** ML can automate decision-making processes and complex computations.
- **Diverse Applications:** ML can be applied to a wide range of domains, including image recognition, natural language processing, fraud detection, and recommendation systems.

Types of Machine Learning

- **Supervised Learning:**

Definition: Models are trained on labeled data, where the outcome is known.

Examples: Regression, classification.

Use Cases: Predicting house prices, classifying emails as spam or not.



- **Unsupervised Learning:**

Definition: Models are trained on unlabeled data, where the outcome is not known.

Examples: Clustering, association.

Use Cases: Customer segmentation, market basket analysis.

- **Semi-Supervised Learning:**

Definition: Models are trained on a combination of labeled and unlabeled data.

Use Cases: Improving model accuracy when labeled data is scarce.

- **Reinforcement Learning:**

Definition: Models learn by interacting with an environment and receiving feedback.

Examples: Q-learning, deep reinforcement learning.

Use Cases: Game playing, robotic control.





Relation Between Machine Learning and Data Science

Data Science is a multidisciplinary field that involves extracting insights and knowledge from data through various techniques, including statistics, data analysis, and machine learning. Machine learning is a critical component of data science, enabling data scientists to build predictive models and automate decision-making processes. Here's how they relate:

- **Data Collection and Preparation:**

Data scientists gather and preprocess data, which is a crucial step before applying machine learning algorithms. This includes cleaning data, handling missing values, and transforming data into suitable formats.

- **Exploratory Data Analysis (EDA):**

Data scientists perform EDA to understand the data's structure, distribution, and relationships. This helps in selecting appropriate machine learning models and features.

- **Feature Engineering:**

Data scientists create new features from raw data to improve model performance. Feature engineering is essential for machine learning as it directly impacts the model's accuracy.



- **Model Building:**

Machine learning models are built and trained on prepared data. Data scientists choose and implement various algorithms (e.g., regression, classification, clustering) based on the problem and data characteristics.

- **Model Evaluation:**

Data scientists evaluate the performance of machine learning models using metrics like accuracy, precision, recall, and F1 score. This ensures the model's reliability and validity.

- **Deployment and Monitoring:**

Once a model is built and validated, data scientists deploy it into production environments. They also monitor the model's performance over time and retrain it as needed.

- **Communication and Visualization:**

Data scientists use data visualization tools to present findings and insights derived from machine learning models to stakeholders. Effective communication is key to making data-driven decisions.